SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	04 July 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
opticrop_vercel_ready/
|-- api/
|   `-- index.py          - Flask app: routes for '/' (input form) and '/predict' (ML
inference)
|-- templates/
|   |-- index.html        - Farmer input form (N, P, K, pH, temperature, humidity,
rainfall)
|   `-- result.html       - Recommendation result page
|-- static/
|   `-- style.css         - Styling for index.html and result.html
|-- model/
|   |-- crop_model.pkl    - Trained scikit-learn crop classifier
|   `-- label_encoder.pkl - Decodes model output into crop names
|-- requirements.txt      - Python dependencies (Flask, numpy, scikit-learn, pandas,
gunicorn)
|-- vercel.json           - Vercel deployment/routing configuration
`-- README_DEPLOY.txt     - Deployment notes and setup steps
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-indol.vercel.app/
Login Credentials (Demo)	Not applicable – the application is openly accessible with no login required
Platform / Hosting Provider	Vercel
Repository Link	https://github.com/1010999/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	https://drive.google.com/file/d/1jh-_2T_EzK96Rc4j7oMxD5ipphJhUWRD/view?usp=drivesdk

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites: Python 3.9+ and pip installed. Step 1: Clone or download the project folder (opticrop_vercel_ready/). Step 2: Install dependencies: pip install -r requirements.txt Step 3: Ensure model/crop_model.pkl and model/label_encoder.pkl are present in the model/ folder. Step 4: Run locally: python api/index.py (or: flask --app api/index.py run) Step 5: Open a browser at http://localhost:5000 (or the port shown in the terminal). To deploy to Vercel instead of running locally: Step 1: Install the Vercel CLI: npm install -g vercel Step 2: From the project folder, run: vercel --prod (vercel.json is already configured to route all requests to api/index.py)

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Cold-start latency on Vercel: max response time reached ~11-15s during load testing, versus ~0.4-1.2s typical/average.	Known limitation of serverless hosting; potential fix is periodic keep-warm pings or moving to an always-on server.
2	No server-side input validation or error handling; malformed or out-of-range input could cause an unhandled error.	Planned improvement – add try/except and range checks in the /predict route.
3	No authentication or user roles; the app is fully open access.	Acceptable for the current single-purpose demo scope; flagged as a future enhancement if multi-user roles are needed.